National Textile University

Department of Computer Science

Subject: Operating System

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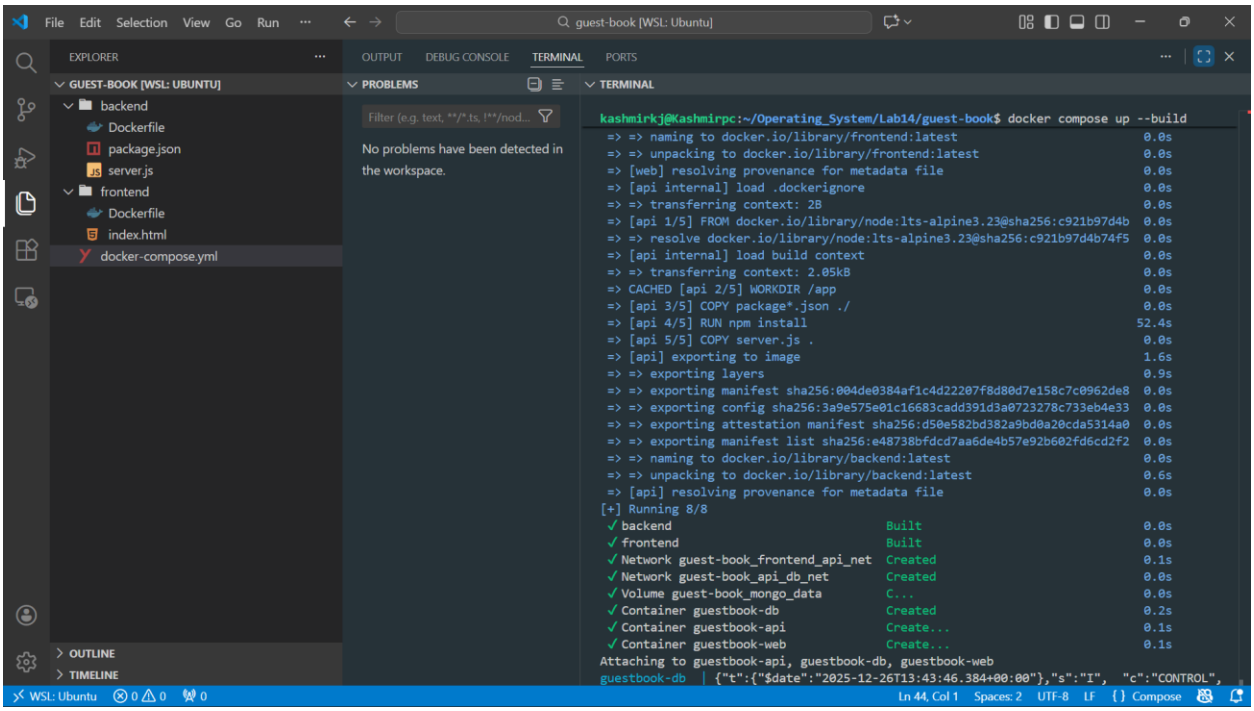
Reg. number: 23-NTU-CS-1167

Lab No 14 (Docker #3)

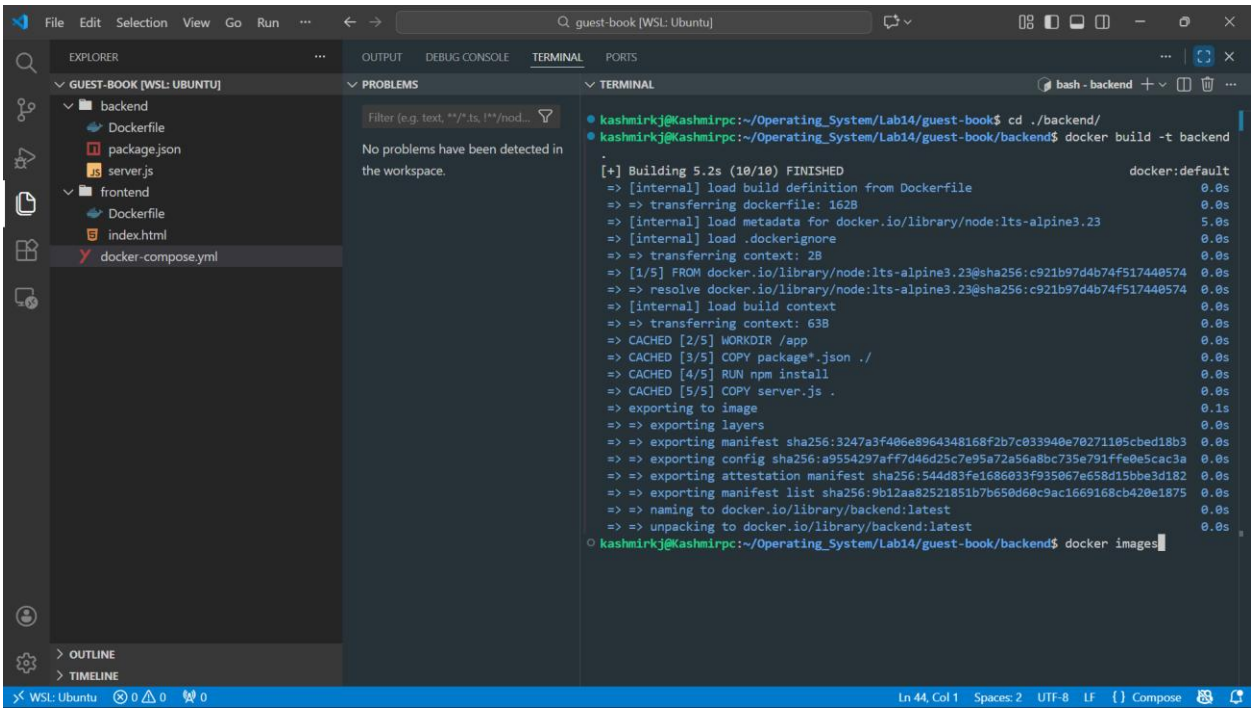
Semester:5th

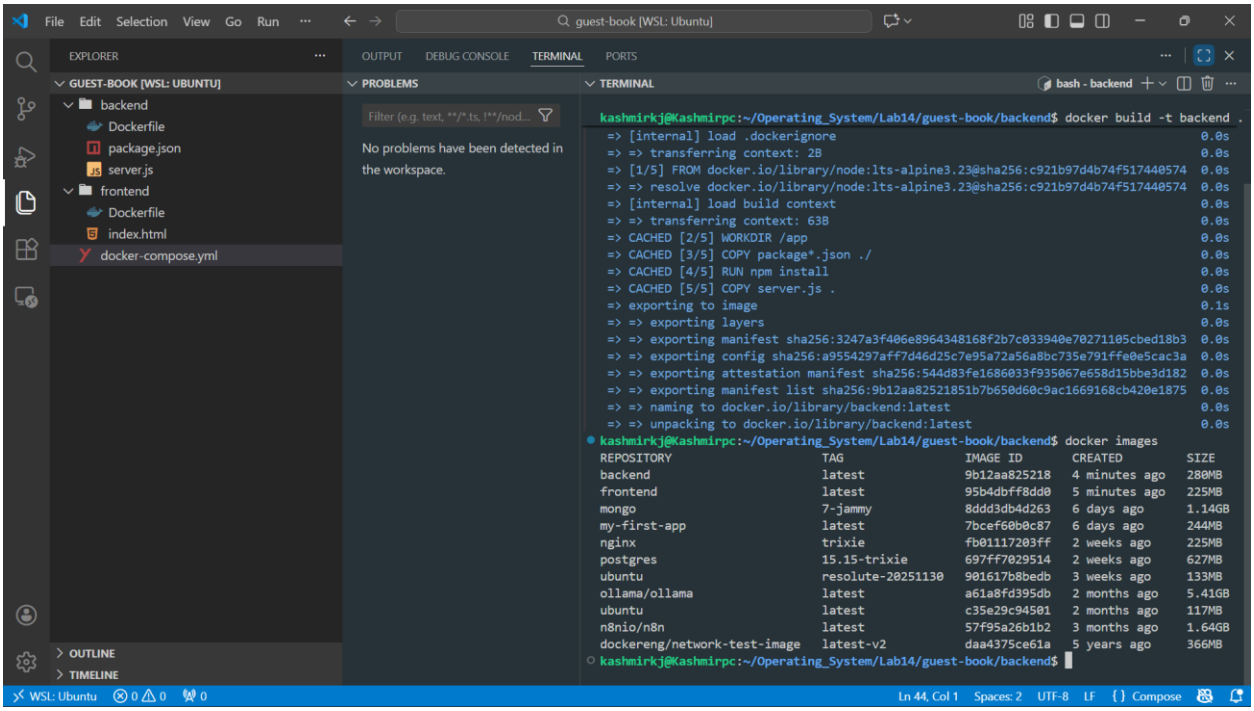
Operating Systems – COC 3071

Whole Structure of project :



Backend Image Creation :



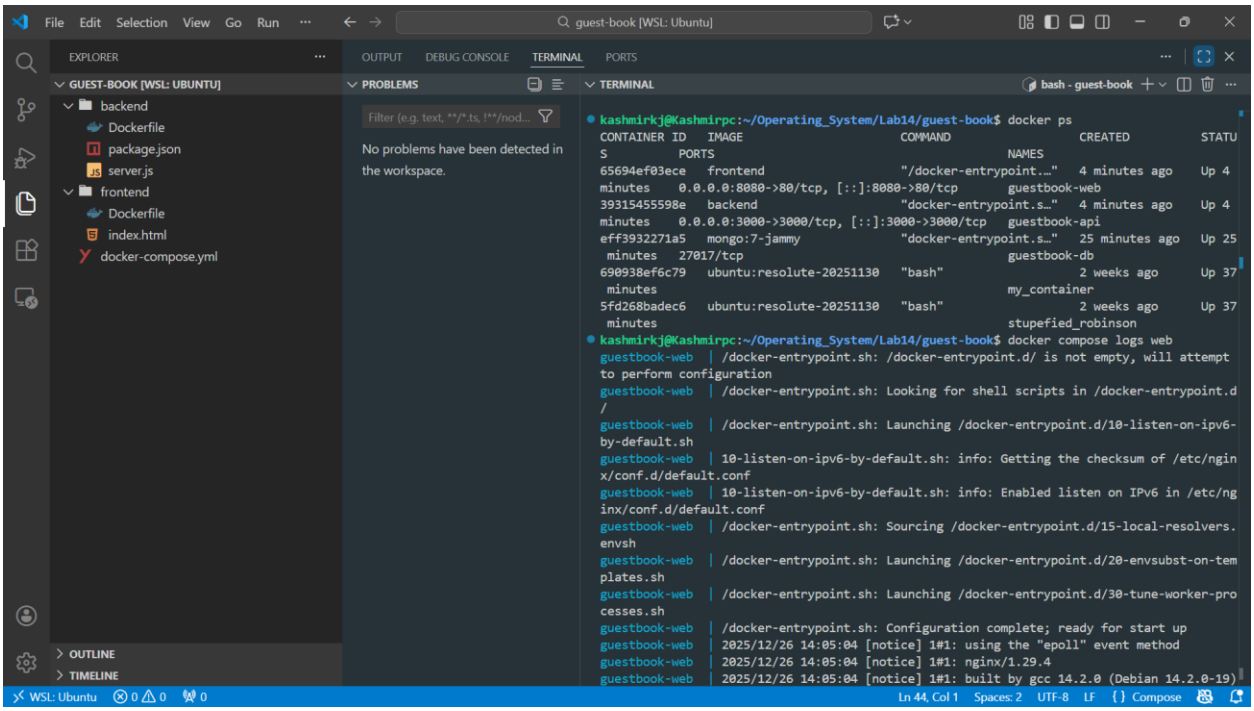


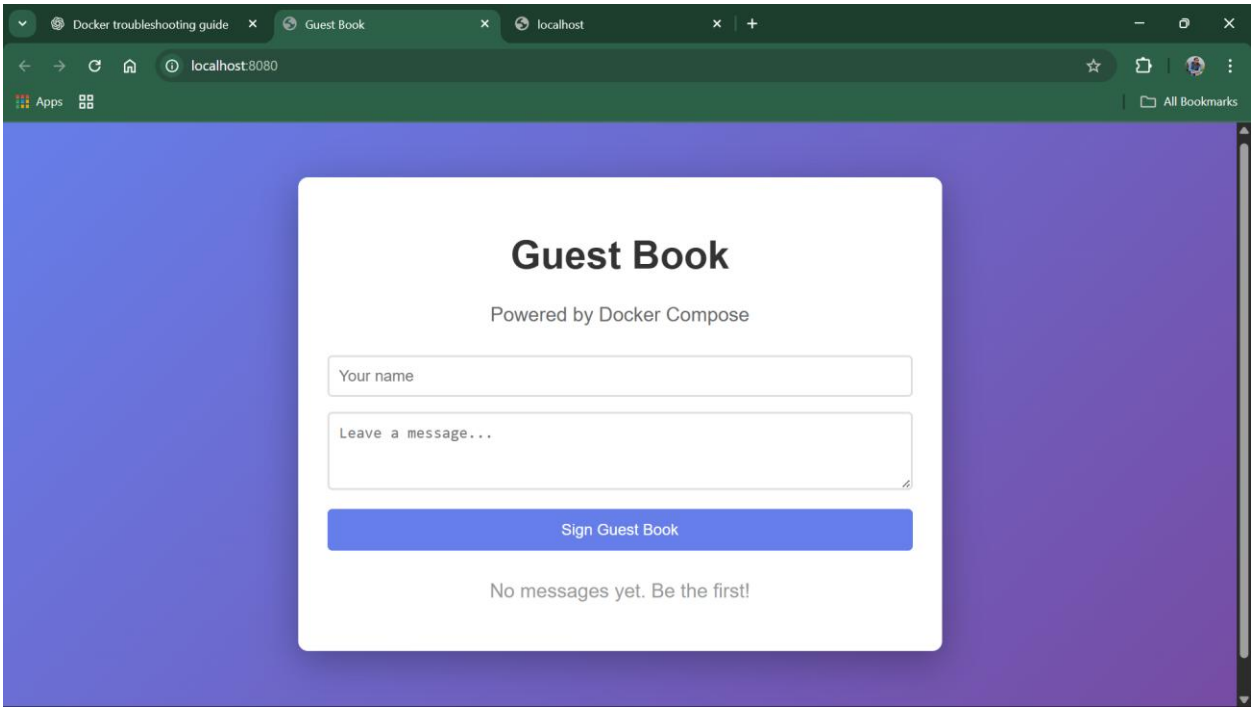
Frontend Image Creation:

```
Run 'docker buildx build --help' for more information
kashmirkj@Kashmirpc:~/Operating_System/Lab14/guest-book/frontend$ docker build -t frontend
.
[+] Building 0.4s (7/7) FINISHED                                    docker:default
 => [internal] load build definition from Dockerfile                          0.0s
 => => transferring dockerfile: 103B                                          0.0s
 => [internal] load metadata for docker.io/library/nginx:trixie               0.1s
 => [internal] load .dockerignore                                             0.0s
 => => transferring context: 2B                                               0.0s
 => [internal] load build context                                             0.0s
 => => transferring context: 32B                                              0.0s
 => [1/2] FROM docker.io/library/nginx:trixie@sha256:fb01117203ff38c2f9af91db1a7409  0.0s
 => => resolve docker.io/library/nginx:trixie@sha256:fb01117203ff38c2f9af91db1a7409  0.0s
 => CACHED [2/2] COPY index.html /usr/share/nginx/html/                       0.0s
 => exporting to image                                                        0.1s
 => => exporting layers                                                       0.0s
 => => exporting manifest sha256:7847eeee7e09efe8c81d54af29ecc5f07af7c7c1c8f4c00df2  0.0s
 => => exporting config sha256:416f892f8fe1b19b46061f56a2ebbd2a9b1cd50916423dce7f9d  0.0s
 => => exporting attestation manifest sha256:5cf86f2c1b8616c948aea87d8ce2f8e35d8aa8  0.0s
 => => exporting manifest list sha256:44e7bb67bd46ba47c522a30dcff3cbaf4d802554ab60b  0.0s
 => => naming to docker.io/library/frontend:latest                            0.0s
 => => unpacking to docker.io/library/frontend:latest                         0.0s
kashmirkj@Kashmirpc:~/Operating_System/Lab14/guest-book/frontend$
```

```
kashmirkj@Kashmirpc:~/Operating_System/Lab14/guest-book/frontend$ docker pull mongo:7-jamm
y
7-jammy: Pulling from library/mongo
Digest: sha256:8ddd3db4d2638eb914cce56284e2f0d6daf140bba31679b2af86f7d790a4c77e
Status: Image is up to date for mongo:7-jammy
docker.io/library/mongo:7-jammy
kashmirkj@Kashmirpc:~/Operating_System/Lab14/guest-book/frontend$
```

Web Page (Frontend):





Code :

Docker -Compse-yml:

```yaml
services:
  # MongoDB Database
  mongodb:
    image: mongo:7-jammy
    container_name: guestbook-db
    volumes:
      - mongo_data:/data/db
    networks:
      - api_db_net

  # Backend API
  api:
    build: ./backend
    image: backend
    container_name: guestbook-api
    environment:
      - MONGO_URL=mongodb://mongodb:27017
    ports:
      - "3000:3000"
    depends_on:
      - mongodb
    networks:
      - frontend_api_net
      - api_db_net

  # Frontend
  web:
    build: ./frontend
    image: frontend
    container_name: guestbook-web
    ports:
      - "8080:80"
    depends_on:
      - api
    networks:
      - frontend_api_net

networks:
  frontend_api_net:
  api_db_net:

volumes:
  mongo_data:

```

Backend Docker File :

```
1  FROM node:lts-alpine3.23
2  WORKDIR /app
3  COPY package*.json ./
4  RUN npm install
5  COPY server.js .
6  EXPOSE 3000
7  CMD ["npm", "start"]
```

Frontend Docker File :


FROM nginx:trixie
COPY index.html /usr/share/nginx/html/
EXPOSE 80